In math, we deal with a lot of theorems, formulas, and properties. For example, we know that the sum of the angles in a triangle is 180 degrees. We assume these to be true because we're taught that they're true, but how do we know?

Well, a long time ago, someone proved that they were true with logic. We start off with postulates (AKA axioms) that we assume to be true because they are self-evident from observation. (This is mostly true for geometry.) We build off of these fundamental postulates. An example of a postulate in Euclidean geometry (named after Euclid) is the postulate that all right angles are equal. This seems pretty obvious, but we can't prove it from just reasoning. Postulates become theorems proved from a chain of reasoning.

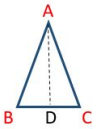
In Math 2, we do proofs using 2 columns. One column gives a statement, and another column gives a justification for why that statement is true.

Two Column Proof: Isosceles Triangle Theorem

The Isosceles Triangle Theorem states that if two sides of a triangle are congruent, then the angles opposite the sides are congruent.

Given: $\overline{AB} \cong \overline{AC}$

Prove: $\angle B \cong \angle C$



Statements	Reasons
1. $\overline{AB} \cong \overline{AC}$	1. Given
2. Construct a bisector of $\angle A$	2. Every $\angle$ has 1 $\angle$ bisector
3. $\angle BAD \cong \angle CAD$	3. Definition of $\angle$ bisector
4. $\overline{AD} \cong \overline{AD}$	4. Reflexive Property
5. $\Delta ABD \cong \Delta ACD$	5. SAS Postulate
6. $\angle B \cong \angle C$	6. CPCTC

In a problem, you will be given statements that are true. From those statements, we want to deduce the truth of other statements.

Even for the simplest of statements, we have to provide a justification.

Example: Prove that A = B when A = 3 and B = 3.

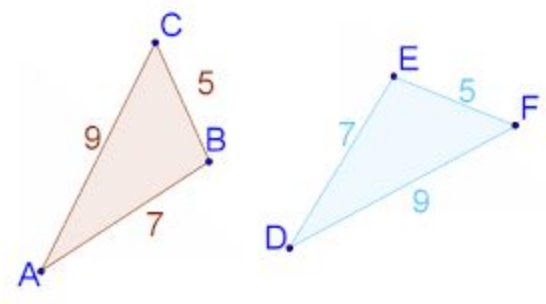
Statements	Reasons
1. A = 3 2. B = 3 3. A = B	1. Given 2. Given 3. Transitive Property

If something is given in the problem, then we have to justify it using the reason "Given". We used the transitive property to substitute B in for 3 in the equation A = 3. Even with the simplest of arithmetic, you have to give a justification. You must also justify that something equals itself with the reflexive property.
Example: Prove that A=A.

Statements	Reasons
1. A = A	Reflexive Property

With the triangle congruence rules we learned, you can prove triangles to be congruent.

Ex: Prove that triangle ABC is congruent to triangle DEF.

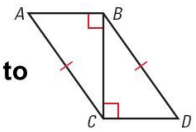


Statements	Reasons
1. AB = DE 2. BC = EF 3. AC = DF 4. ABC is congruent to DEF	1. Given 2. Given 3. Given 4. SSS

Practice problems

Use the diagram at the right.

4. Use the information in the diagram to prove that $\triangle ACB \cong \triangle DBC$



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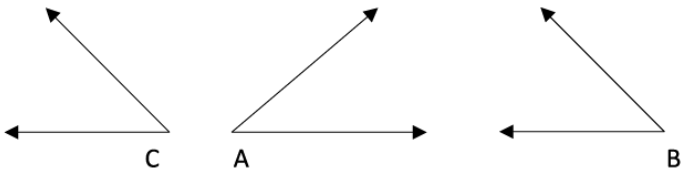
Given: $\angle A$ and $\angle C$ are complementary
$\angle B$ and $\angle C$ are complementary

Prove: $\angle A \cong \angle B$



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